

In-vehicle communication systems

In-vehicle communication systems (IVCS) are just like networks in the world of computers. The key difference is that in the case of IVCS, the communication is typically restricted within a vehicle, and is between different components of the vehicle; for example, the sensors within a fuel tank communicating with the digital display on the dashboard.

For in-vehicle communications, many different protocols have been developed, popular ones being the controller area network (CAN) and the FlexRay.

Controller area network. Due to its speed of 1 Mbps, the controller area network is often used as the conduit to transmit critical information to the main controller in the automobiles. It is used in cars for enabling robust serial communication, in order to make them more reliable, safe and fuel-efficient while decreasing the wiring complexity. It is just like any other networking protocol which follows a layered approach to system implementation. The layered approach allows interoperability between products from different manufacturers.

R. Natarajan, vice president-automotive and aerospace practice, Tata Elxsi, explains, "Besides CAN interface, the other technologies used in the automotive industry are LIN, FlexRay, Class2, Kline, MOST, SAE J1850, J1939 and Keyword2000, to name a few. Tata Elxsi is an associate member of the FlexRay Consortium."

FlexRay. An alternative to the CAN, FlexRay operates at ten times faster speeds. It uses two channels for communication with an embedded redundancy mechanism. It is also time-deterministic, which means that it ensures information will be sent accurately and quickly.

Desmond says, "In the control LAN environment, Fujitsu has been an associate member of the FlexRay Consortium started in 2000 and is instrumental

Future Technologies

➤ **Blind-detection spot alerts**

New models from General Motors, Mazda and Lexus have built-in blind-detection spot alerts.

➤ **Vehicle slowdown system**

GM's vehicle slowdown system puts your car to an idle speed and switches on the emergency flashers to enable vehicle recovery. If you want to find a restaurant, tell its navigation system "restaurant; 4 star; Thai food" and it will not only locate your dinner but route you there.

➤ **Traffic and weather alert navigation**

In the future, drivers will be more informed about weather and traffic conditions with real-time information superimposed on GPS-based navigation systems. The weather or traffic information is scaled and customised for the driver's location and route. The service provides alerts about severe traffic and weather conditions that can help drivers avoid a massive traffic jam or a storm brewing on the road ahead.

➤ **Adaptive headlamps**

A vehicle's headlamps can now follow wherever the driver turns the steering wheel, to illuminate where the driver is looking instead of the car's front end only. The adaptive headlamps system involves processing of information about the location of the steering wheel and the speed of the vehicle, which then signals electric motors behind the headlamps to turn left or right accordingly.

➤ **Collision avoidance system**

The collision avoidance system monitors the distance between the cars or if any obstacle is ahead of the car. In the case of obstacle, or if the car comes very close to another, the system alerts the driver with flashing light, vibrating seat, tugging seat belt and more. In some cases when the vehicle is very close to another car, the system itself applies the brakes.

in the development of this technology. FlexRay is poised to replace CAN as the technology of choice. Recently, FlexRay technology has been implemented in BMW's X5 series."

Others. Besides FlexRay, there are other products developed based on IDB1394 standards. Desmond comments, "IDB1394 is suitable for transmission of digital audio-video in the automobile environment as it supports 400Mbps transmission speed, which is enough to allow multiple high-definition video and audio streams to be transported and shared between the passengers in the car."

He adds, "IDB1394 will help to shed a hefty load of copper harness (about 20+ kg) off the car. This load saving will translate into higher mileage for the same amount of gasoline used and contribute to lowering of CO₂ emission and environment protection."

The future

In the future, we will see introduction of many other new features in cars.

Cars are becoming better and better every year.

Chetan Maini, deputy chairman and chief technology officer at REVA Electric Car Company, says, "All the trends in the automotive industry point to a convergence of the automotive and electronics industries that is literally transforming the automobiles. Soon cars will be able to receive signals from other vehicles and then use computerised controls to take actions such as slowing to avoid collisions. As many experts in the industry feel, new cars already contain more electronics than steel."

Maini adds, "GPS, satellite navigation, wireless communications, spatial detection devices and parking sensors are being integrated into cars. There are also technologies being integrated into cars that automatically summon help in the event of crashes and pinpoint the locations of stolen vehicles." Another category of car that has made its way in the market is the hybrid car. ●

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